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Anomalies of the Upper Urinary Tract

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Keywords

crossed fused ectopic; ectopic kidney; horseshoe kidney; renal agenesis; renal artery aneurysm

Questions

1. During a left inguinal herniorrhaphy, the vas deferens is absent and a 3-mm golden-yellow nodule is found along the spermatic cord. This boy may also have:
    a. a left appendage epididymis.
    b. left renal agenesis.
    c. malpositioned left adrenal gland.
    d. absent left head of epididymis.
    e. absent left testis.
2. A 14-year-old girl with abdominal pain undergoes an abdominal and pelvic ultrasound. A solitary right kidney is seen. Her abdominal pain is most likely associated with:
    a. skeletal anomalies.
    b. a unicornuate uterus.
    c. imperforate hymen.
    d. a didelphic uterus.
    e. an absent left ovary.

3. The most frequent renal fusion anomaly is:
    a. pancake kidney.
    b. crossed fused renal ectopia.
    c. horseshoe kidney.
    d. disc kidney.
    e. Sigmoid kidney.
4. The renal segment with the most variable blood supply is the:
    a. apex.
    b. upper.
    c. middle.
    d. lower.
    e. posterior.
5. A 22-year-old hypertensive woman is found to have a 2.7-cm renal artery aneurysm (RAA). Treatment is recommended:
    a. if the RAA rapidly decreases in size.
    b. when the woman is no longer of childbearing age.
    c. at this time.
    d. if there is no flank pain from emboli originating in the RAA.
    e. when the hypertension is well controlled.
6. A 5-year-old girl with a pelvic kidney has hydronephrosis most commonly due to:
    a. vesicoureteral reflux.
    b. malrotation.
    c. ureterovesical junction obstruction.
    d. ureteropelvic junction obstruction.
    e. ectopic ureter.
7. A newborn girl was noted prenatally to have coarctation of the aorta and a horseshoe kidney. After a renal bladder ultrasound, the next step is to obtain a(n):
    a. voiding cystourethrogram.
    b. magnetic resonance urography (MRU) with gadolinium.
    c. echocardiogram.
    d. karyotype.
    e. skeletal series.
8. Unilateral renal agenesis is commonly associated with:
    a. normal position of the splenic flexure.

b. normal position of the adrenal gland.
c. ipsilateral undescended testis.
d. normal position of the hepatic flexure.
e. rudimentary uterus.

9. A 2-month-old male is seen for prenatal ultrasound finding of left renal agenesis, and this is confirmed with postnatal ultrasound. The right kidney ultrasound findings show normal renal echogenicity and no renal collecting system or ureteral dilation. What testing/counseling would be advised?
   a. A voiding cystourethrogram (VCUG) must be performed.
   b. Serial renal ultrasounds annually for life even if asymptomatic.
   c. A confirmatory diuretic renal scan.
   d. Lifelong blood pressure check and urine analysis to monitor for evidence of chronic kidney disease.
   e. Nuclear glomerular filtration rate measurement.
10. Unilateral renal agenesis (URA) is best reliably diagnosed by finding:
   a. a single umbilical artery.
   b. preauricular skin tag(s).
   c. an imperforate hymen.
   d. absence of renal artery at L1-L2.
   e. specific radiographic evidence.
11. Male predominance of occurrence is most striking in:
   a. unilateral renal agenesis.
   b. bilateral renal agenesis.
   c. crossed fused renal ectopia.
   d. ectopic kidney.
   e. calyceal diverticulum.
12. The incidence of unilateral renal agenesis is:
   a. 1:2500.
   b. 1:4000.
   c. 1:1100.
   d. 1:5000.
   e. 1:500.

13. Unilateral renal agenesis and a unicornuate uterus will form when the embryologic insult occurs at which gestational time?
    a. Before the fourth week
    b. At the start of the fourth week
    c. At the end of the fourth week
    d. At the start of the fifth week
    e. At the end of the fifth week
14. In autopsy studies, unilateral renal agenesis was found in association with:
    a. absence of the gonad.
    b. a normally developed ureter.
    c. an ectopic ureteral orifice.
    d. adrenal agenesis.
    e. absence of the head of the epididymis.
15. Most ectopic kidneys are clinically asymptomatic EXCEPT:
    a. pelvic kidneys.
    b. thoracic kidneys.
    c. kidneys with ectopic ureters.
    d. lumbar kidneys.
    e. abdominal kidneys.
16. The isthmus of a horseshoe kidney is located adjacent to which vertebrae?
    a. T12 and L1
    b. L1 and L2
    c. L3 and L4
    d. L5 and S1
    e. S1 and S2
17. Between the sixth and ninth week, normal rotation of the kidney toward the midline to attain its orthotopic position involves:
    a. 60 degrees of lateral rotation.
    b. 90 degrees of lateral rotation.
    c. 180 degrees of lateral rotation.
    d. 90 degrees of medial rotation.
    e. 180 degrees of medial rotation.
18. Congenital renal arteriovenous fistulas are:
    a. usually congenital.

b. cirsoid in configuration.
c. symptomatic before the third decade.
d. more common in males.
e. usually located in the lower pole.

19. Bilateral megacalycosis:
    a. occurs more frequently in females.
    b. has an increased number of dilated calyces.
    c. is associated with ureteral dilation.
    d. is autosomal recessive in inheritance pattern.
    e. shows an obstructive pattern on renal scan.

Answers

1. **b. Left renal agenesis.** The finding of a 3-mm golden-yellow nodule is indicative of ectopic adrenal. The adrenal develops just medial to the gonadal ridge. Their close proximity explains their location along the spermatic cord and their incidental identification at the time of herniorrhaphy or orchidopexy. Histologically, the nodules contain adrenal cortex but no medulla. In this case, the absent vas should raise a red flag for possible ipsilateral renal agenesis because the ureteral bud and vas are both derived from the wolffian duct. **In one study, 79% of adult males with absence of the vas deferens have an absent ipsilateral kidney, with left-sided lesions predominating.** The lower pole and mid-pole of the epididymis are wolffian duct derivatives. The head of the epididymis is derived from the mesonephric tubules, which link the mesonephric or wolffian duct with the gonad.
2. **d. A didelphic uterus.** Unilateral renal agenesis can be associated with didelphic uterus and obstruction of the ipsilateral vagina resulting in hematocolpos. This syndrome is referred to as OHVIRA (obstructed hemivagina ipsilateral renal agenesis). This would likely explain this girl's abdominal pain.
3. **c. Horseshoe kidney.** This anomaly represents the most frequently observed renal fusion anomaly.

4. **a. Apex.** The vessel to the apical segment has the greatest variation in origin; it arises from (1) the anterior division (43%), (2) the junction of the anterior and posterior divisions (23%), (3) the main-stem renal artery or aorta (23%), or (4) the posterior division of the main renal artery (10%).
5. **c. At this time. Generally, treatment is recommended if patient presents with RAA rupture or deemed to be at high risk for rupture of the RAA. High risk for rupture is associated with rapidly expanding RAA or in pregnant females and those females who are considering pregnancy. Uncontrollable hypertension in association with renal artery stenosis and "shower" emboli from the RAA are additional indications.**
6. **d. Ureteropelvic junction obstruction. The renal pelvis is usually anterior (instead of medial) to the parenchyma because the kidney has incompletely rotated. As a result, 56% of ectopic kidneys have a hydronephrotic collecting system.** Half of these cases are due to obstruction of the ureteropelvic or the ureterovesical junction (70% and 30%, respectively), 25% from reflux grade 3 or greater, and 25% from the malrotation alone.
7. **d. Karyotype.** Horseshoe kidney and coarctation of the aorta are seen in patients with Turner syndrome (45,XO). Therefore, a karyotype should be obtained. Other stigmata may include lymphedema, shield chest, low hairline, and webbed neck.
8. **b. Normal position of the adrenal gland.** Unilateral renal agenesis is commonly associated with an adrenal gland that is in a normal position, although it may be flattened. Regardless of sex, both gonads are usually normal. **The most common müllerian duct anomalies are a true unicornuate uterus with complete absence of the ipsilateral horn and fallopian tube or a bicornuate uterus with rudimentary development of the horn on the affected side.** A plain film of the abdomen (or other radiographic study such as magnetic resonance imaging) showing the gas pattern of the splenic flexure in the left renal fossa suggests left renal agenesis, ectopia, or crossed ectopia, whereas the gas pattern of the hepatic flexure positioned in the right renal fossa suggests congenital absence of the right kidney.

9. **d. Lifelong blood pressure check and urine analysis to monitor for evidence of chronic kidney disease.** Given the increased rate of chronic kidney disease seen in children with solitary functioning kidney, current suggestions include baseline and annual surveillance assessments of blood pressure and urine studies for proteinuria because hypertension and microalbuminuria are features of progressive decrease in glomerular filtration rate. A VCUG may reveal reflux at higher rate in unilateral renal agenesis but it is not a mandatory test. Similarly, a well-performed renal ultrasound that does not reveal an ectopic kidney makes a renal scan to confirm the diagnosis an elective decision rather than mandatory one. There is no documented benefit to perpetual lifelong ultrasound screening in unilateral renal agenesis.
10. **e. Specific radiographic evidence.** Unilateral renal agenesis can be diagnosed reliably with radiographic examinations including abdominal and pelvic ultrasound, dimercaptosuccinic acid (DMSA) scan, and/or magnetic resonance angiography (MRA).
11. **b. Bilateral renal agenesis.** Male predominance is most striking in bilateral renal agenesis, with almost 75% of affected individuals being male. For unilateral renal agenesis, there is a male-to-female ratio of 1.8:1. Crossed fused ectopia has a slight male predominance (3:2), whereas ectopic kidneys have no significant difference in incidence between the sexes.
12. **c. 1:1100.** The incidence of unilateral renal agenesis is 1:1100.
13. **a. Before the fourth week.** Unilateral renal agenesis and a unicornuate uterus will form when the embryologic insult occurs before the fourth week. If the insult occurs early in the fourth week of gestation and affects both the wolffian duct and the ureteral bud, maldevelopment of the wolffian duct affects renal development, müllerian duct elongation, contact with the urogenital sinus, and subsequent fusion. Therefore, a didelphic uterus will form with obstruction of the horn and vagina on the side of the unilateral renal agenesis. If the insult occurs after the fourth week, the wolffian duct and müllerian duct elongation and differentiation proceed normally and only the ureteral bud and metanephric

blastema are affected, thereby resulting in isolated unilateral renal agenesis.

14. **d. Adrenal agenesis.** In autopsy studies of unilateral renal agenesis, adrenal agenesis occurs in fewer than 10%, although the ipsilateral adrenal gland may be flattened or "lying down." The ureter is not normally developed, and the ipsilateral ureter is completely absent in approximately 60% of cases. The gonad is usually normal in both sexes. The head of the epididymis is normally formed because it is derived from the mesonephric tubules that link the mesonephric duct to the gonad.
15. **c. Kidneys with ectopic ureters.** Most ectopic kidneys are clinically asymptomatic except for the unusual cases of an ectopic kidney with an ectopic ureter.
16. **c. L3 and L4.** The isthmus of a horseshoe kidney is located adjacent to the third and fourth lumbar vertebrae.
17. **d. 90 degrees of medial rotation.** Between the sixth and ninth week, normal rotation of the kidney toward the midline to attain its orthotopic position involves 90 degrees of medial rotation.
18. **b. Cirsoid in configuration.** Fewer than 25% of all renal arteriovenous fistulas (AVFs) are congenital. They are identifiable by their cirsoid configuration and multiple communications between the main or segmental renal arteries and venous channels. Although congenital, they rarely present clinically before the third or fourth decade. Women are affected three times as often as men, and the right kidney is involved slightly more often than the left. The lesion is usually located in the upper pole (45% of cases), but not infrequently it may be found in the mid-portion (30%) or in the lower pole (25%) of the kidney.
19. **b. Has an increased number of dilated calyces.** Megacalycosis is defined as a nonobstructive enlargement of calyces resulting from malformation of the renal papillae. The calyces are generally dilated and malformed and may be increased in number. The renal pelvis is not dilated, nor is its wall thickened, and the ureteropelvic junction is normally funneled without evidence of obstruction. The ureter is usually normal. It occurs predominantly in males in a ratio

of 6:1. Bilateral disease has been seen almost exclusively in males, whereas segmental unilateral involvement occurs only in females.

Chapter Review

1. In bilateral renal agenesis, 40% of affected infants are stillborn. The ureters are almost always absent, and the bladder is either absent or hypoplastic. The adrenal glands, however, are usually in their normal anatomic position.
2. In patients with bilateral renal agenesis associated with oligohydramnios, Potter facies are pathognomonic of the process. Pulmonary hypoplasia is frequently present.
3. Ultrasound screening is recommended for parents and siblings of infants with unilateral or bilateral renal agenesis—there is a higher risk of renal agenesis in this population.
4. In unilateral renal agenesis, the ipsilateral ureter is completely absent in 60% of cases. Abnormalities of the contralateral ureter are not uncommon; reproductive tract anomalies in females are also common.
5. With unilateral renal agenesis, 25% of the contralateral ureters reflux.
6. There is an association of genital anomalies with renal ectopia. The upper pole of the ectopic kidney usually joins with the lower pole of the normal kidney.
7. In all types of fusion anomalies, the ureter from each kidney is usually orthotopic.
8. The highest incidence of associated anomalies occurs with solitary renal ectopia. Associated anomalies in the male include cryptorchidism, and vaginal atresia or unilateral uterine anomalies in the female.
9. In a horseshoe kidney, the isthmus is bulky and consists of parenchymatous tissue with its own blood supply. The blood supply to a horseshoe kidney is variable.
10. Ureteropelvic junction obstruction in horseshoe kidneys occurs one-third of the time.

11. The incidence of Wilms tumors and renal pelvic tumors in horseshoe kidneys is higher than would be expected in the general population. There is no increased risk of renal cell carcinoma.
12. Renal arteries are end arteries and, as such, have no collaterals.
13. Arteriovenous fistulas may result in hypertension in 50% of cases, due to relative ischemia beyond the fistula. It is renin-mediated hypertension.
14. Infundibulopelvic stenosis is usually bilateral and is commonly associated with vesicoureteral reflux.
15. Maternal diabetes is associated with a threefold increased risk of renal agenesis and dysplasia.
16. Patients born with renal agenesis may have decreased renal reserve in the remaining kidney, which results in a significant risk for end-stage renal disease.
17. Of adult males with absence of the vas deferens, 79% have an absent ipsilateral kidney.
18. Excision of a renal artery aneurysm is recommended if (1) the hypertension cannot be easily controlled; (2) incomplete ring-like calcification is present; (3) the aneurysm is larger than 2.5 cm; (4) the patient is female and of childbearing age, because rupture during pregnancy is a likely possibility; (5) the aneurysm increases in size on serial angiograms; or (6) an arteriovenous fistula is present.

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Renal Dysgenesis and Cystic Disease of the Kidney

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Keywords

cystic lesion; kidney; renal dysgenesis

Questions

1. Which of the following is a correct match regarding renal anomaly and associated finding?
    a. von Hippel-Lindau disease and adenoma sebaceum
    b. Tuberous sclerosis and angiomyolipoma
    c. Autosomal dominant polycystic kidney disease (ADPKD) and salt-losing nephropathy
    d. Congenital nephrosis (Finnish type) and medullary cysts
    e. Autosomal recessive polycystic kidney disease (ARPKD) and colonic diverticulosis
2. The primary feature(s) associated with Ask-Upmark kidney (segmental hypoplasia) is/are:
    a. hypertension.
    b. renal artery intimal disease.
    c. found in young men and boys.
    d. b and c.
    e. a and c.

3. The development of acquired renal cystic disease (ARCD) is most related to which factor?
    a. Age of the patient
    b. Duration of renal failure
    c. Recent initiation of hemodialysis
    d. *Escherichia coli* infection
    e. Genetic defect on chromosome 16
4. Which statement(s) about ARPKD is/are TRUE?
    a. The most severe forms develop in late childhood or adolescence.
    b. No matter the severity of the renal disease, all patients will have liver involvement in the form of congenital hepatic fibrosis.
    c. In newborns, ultrasound findings include very enlarged kidneys with increased parenchymal echogenicity.
    d. a and b.
    e. b and c.
5. Which of the following statements accurately describes a fundamental process essential for the development of renal cysts?
    a. Proliferation of epithelial cells in segments of the renal collecting system
    b. Accumulation of fluid within an expanding segment of the glomerulus
    c. An imbalance of the secretory and absorptive properties in proliferating tubular epithelial cells
    d. Hypertrophy of the basement membrane within the ascending loop of Henle
    e. Glomerular outpouching resulting from elevated glomerular hydrostatic pressure
6. Which of the following statement(s) is/are correct about ADPKD?
    a. The genetic defect is located on the short arm of chromosome 16.
    b. Most affected infants have congenital hepatic fibrosis.
    c. Renal cysts are infrequently seen on ultrasound in affected patients before 30 years of age.

d. Glomerular cysts are never found in the kidneys of newborns diagnosed with ADPKD.
e. The incidence of renal cell carcinoma in ADPKD is twice that in the normal population.

7. All of the following are extrarenal manifestations of ADPKD EXCEPT:
   a. hepatic cysts.
   b. intracranial (berry) aneurysms.
   c. cerebellar hemangioblastomas.
   d. colonic diverticulosis.
   e. mitral valve prolapse.
8. Which of the following statements is FALSE regarding unilateral multicystic dysplastic kidneys?
   a. The majority of multicystic dysplastic kidneys become smaller or ultrasonographically undetectable with time.
   b. There is an absence of communication between cysts on ultrasonographic scans.
   c. Cysts are usually found in communication with each other when injected intracystically with contrast material.
   d. The sine qua non for histologic diagnosis of a multicystic dysplastic kidney is the presence of primitive ducts.
   e. Multicystic dysplastic kidneys appear more often in females and more often on the right side.
9. Flank pain is one of the most common presenting symptoms of ADPKD in adult patients. This is often caused by:
   a. bleeding into a cyst.
   b. renal cell carcinoma.
   c. cyst rupture.
   d. b and c.
   e. a and c.
10. Which gene is associated with clear cell renal cell carcinoma and a multiple malformation syndrome?
   a. PDK1
   b. PDK2
   c. TG737
   d. Wnt-2

e. VHL

11. A benign multilocular cyst is seen most often:
    a. in males younger than 4 years and in females older than 30 years.
    b. in females younger than 4 years and in males older than 30 years.
    c. in males between 4 and 30 years.
    d. equally in both sexes before 4 years and in females after 30 years.
    e. equally in both sexes before 4 years and in males after 30 years.
12. What is the primary distinguishing factor between juvenile nephronophthisis (NPH) and medullary cystic kidney disease (MCKD)?
    a. NPH presents with polyuria and polydipsia, whereas MCKD does not.
    b. NPH is an autosomal recessive disorder, whereas MCKD is an autosomal dominant disease.
    c. NPH is diagnosed histologically with severe interstitial fibrosis, whereas MCKD is diagnosed by the presence of glomerulosclerosis.
    d. Most patients with MCKD have extrarenal manifestations of the disease, whereas patients with NPH are usually affected only in the kidneys.
    e. In patients with NPH, renal failure occurs in the third to fourth decade, whereas in patients with MCKD, renal failure typically occurs in adolescence.
13. A patient with which of the following entities has the highest likelihood of having a renal cell carcinoma develop?
    a. ADPKD
    b. Tuberous sclerosis
    c. von Hippel-Lindau disease
    d. Acquired renal cystic disease
    e. Medullary sponge kidney
14. Which of the following is FALSE pertaining to MCDK?

a. MCDK is one of the most common causes of an abdominal mass in the newborn.
b. In patients with MCDK, the contralateral renal moiety is frequently affected by urologic disease.
c. MCDK is often difficult to differentiate from severe ureteropelvic junction obstruction.
d. Data from large series show that MCDK is associated with an increased risk for hypertension.
e. Roughly 40% of MCDKs will spontaneously involute over time.

15. Which of the following would confirm the diagnosis of tuberous sclerosis?
    a. Renal angiomyolipoma and multiple renal cysts
    b. Hamartomatous rectal polyps and facial adenoma sebaceum
    c. Renal angiomyolipoma and cardiac rhabdomyoma
    d. Multiple renal cysts, hepatic fibrosis, and pheochromocytoma
    e. Mitral valve prolapse, renal angiolipoma, and gingival fibromas
16. The following are true of von Hippel-Lindau (VHL) disease EXCEPT:
    a. VHL disease is an autosomal dominant syndrome.
    b. VHL disease is caused by a mutation in the tumor suppressor gene, VHL, located on chromosome 3.
    c. epididymal cysts are not infrequent in patients with VHL disease.
    d. pheochromocytomas, cerebellar hemangioblastomas, and retinal angiomas are common extrarenal manifestations of VHL disease.
    e. renal cell carcinomas, the most common manifestation, are seen in the vast majority of patients.
17. Renal sinus cysts are most likely derived from:
    a. vascular elements.
    b. renal parenchyma.
    c. renal pelvis.
    d. lymphatic system.
    e. nephrogenic rests.

18. Most simple renal cysts that are identified in utero:
    a. represent the first sign of a multicystic kidney.
    b. represent the first sign of ARPKD.
    c. represent the first sign of ADPKD.
    d. represent a calyceal diverticulum.
    e. resolve before birth.
19. Approximately what percentage of individuals older than 60 years will have an identifiable renal cyst on computed tomography (CT)?
    a. 1% to 5%
    b. 10%
    c. 33%
    d. 75%
    e. 90%
20. Which of the following groups of antibiotics include the best choice for treating an infected renal cyst in a patient with ADPKD?
    a. Trimethoprim-sulfamethoxazole, chloramphenicol, fluoroquinolones
    b. Cephalosporins, trimethoprim-sulfamethoxazole, doxycycline
    c. Gentamicin, cephalosporins, vancomycin
    d. Fluoroquinolones, metronidazole, vancomycin
    e. Doxycycline, amoxicillin, gentamicin
21. All of the following are reasonable treatment strategies for patients with ADPKD EXCEPT:
    a. management of hypertension.
    b. avoidance of surgical treatment for large or multiple cysts in patients with chronic flank pain.
    c. surgical treatment of symptomatic urinary stone disease.
    d. use of lipophilic antibiotics for treatment of a suspected renal cyst infection.
    e. screening with magnetic resonance imaging (MRI) or CT for berry aneurysms in patients with a family history of subarachnoid hemorrhage.
22. In neonates with a unilateral multicystic kidney, what is the incidence of contralateral vesicoureteral reflux?
    a. 0% to 7%
    b. 18% to 43%

c. 50% to 67%
d. 75%
e. 7% to 15%

23. What is the most likely cause of flank pain and hematuria in a 50-year-old patient with end-stage renal disease who has been undergoing dialysis for 5 years?
    a. Acute renal vein thrombosis
    b. Acute renal artery thrombosis
    c. Renal cell carcinoma
    d. ARCD
    e. Uric acid stones
24. Which group of three findings best describes the typical ultrasonographic image of a multicystic dysplastic kidney?
    a. The cysts are organized around a central large cyst; there is no identifiable renal sinus; and there are communications between the cysts.
    b. The cysts have a haphazard distribution; there is absence of a central or medial large cyst; and there are no obvious communications between the cysts.
    c. The cysts have a haphazard distribution; there is no obvious renal sinus; and there is a large central cyst.
    d. Connections exist between the cysts; a medial cyst is present; and a renal sinus is usually present.
    e. The cysts are organized at the periphery; the largest is the central one; and there is an identifiable renal sinus.
25. Which one of the following conditions is most representative of a neoplastic growth?
    a. Benign multilocular cyst
    b. Oligomeganephronia
    c. Multicystic dysplastic kidney
    d. Calyceal diverticulum
    e. Ask-Upmark kidney
26. Which of the following is the best match?
    a. ARPKD and congenital hepatic fibrosis
    b. Medullary sponge kidney and predominance of glomerular cysts

c. Juvenile nephronophthisis and cortical cysts
d. Ask-Upmark kidney and hypotension
e. von Hippel-Lindau disease and adenoma sebaceum

27. Which of the following matches is correct?
    a. ARPKD and chromosome 2
    b. ADPKD and chromosomes 4 and 16
    c. Tuberous sclerosis and chromosomes 9 and 15
    d. von Hippel-Lindau disease and chromosome 4
    e. Juvenile nephronophthisis and chromosome 6
28. A renal cyst with increased number of septa and prominent calcification in a nonenhancing cyst wall does not require exploration, but requires timely follow-up. According to the Bosniak grading system, this cyst would be categorized as:
    a. I.
    b. II.
    c. II F.
    d. III.
    e. IV.
29. Ultrasonography in neonates with ARPKD reveals kidneys that are hyperechogenic or "bright" in appearance. This finding is due to:
    a. the presence of many small punctate calcifications within the renal papillae.
    b. dysplastic, diseased renal parenchyma.
    c. a vast increase in small fat deposits within the renal sinuses.
    d. the presence of numerous microcysts created by tightly compacted, dilated collecting ducts that result in innumerable ultrasonographic interfaces.
    e. the presence of renal hamartomas with increased cortical vascularity.
30. Ultrasound and/or CT criteria for the diagnosis of a simple renal cyst include all the following EXCEPT:
    a. sharp, thin, distinct smooth walls and margins.
    b. thickness of cyst wall less than or equal to 3 mm.
    c. acoustic enhancement behind cyst (ultrasound).
    d. spherical or ovoid shape.
    e. homogeneous with absence of internal echoes.

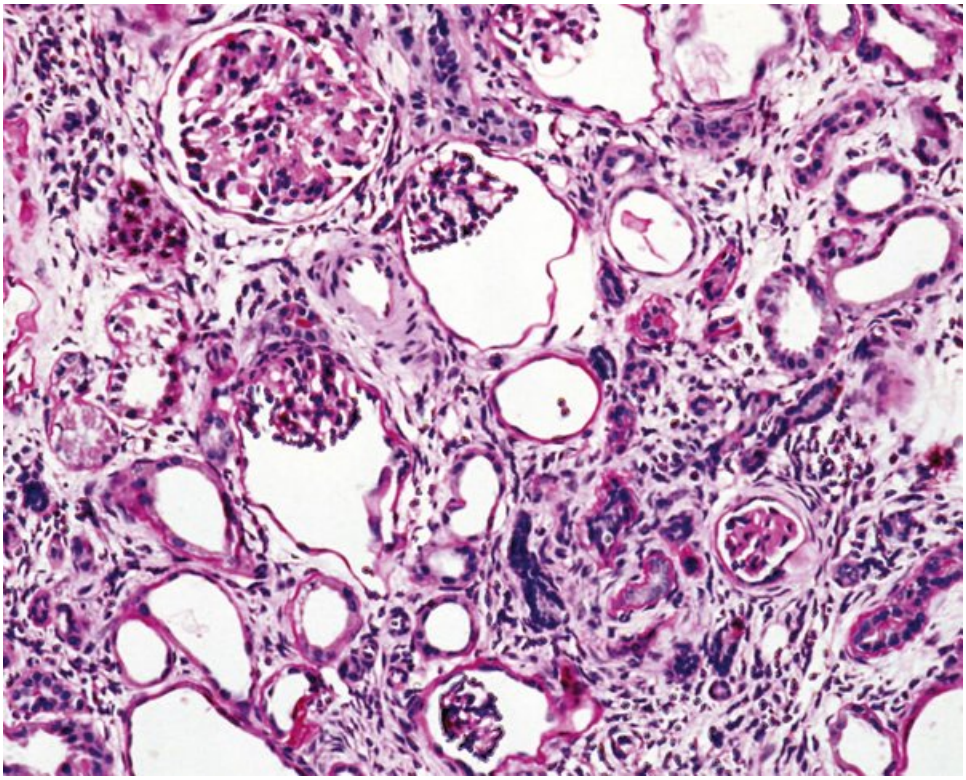


FIG. 39.1
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 3, St. Louis, 2014, Elsevier.

31. A 50-year-old man with known von Hippel-Lindau disease presents with a single episode of gross hematuria. CT scan reveals a 3-cm enhancing mass in the upper pole of each kidney. Metastatic evaluation is negative. He is otherwise healthy. Appropriate treatment at this point would be:
    a. bilateral radical nephrectomy with the placement of a peritoneal dialysis catheter.
    b. bilateral upper pole partial nephrectomy.
    c. right radical nephrectomy with left upper pole partial nephrectomy.
    d. observation with serial CT every 4 months.
    e. CT-guided needle biopsy of each lesion with surgical removal if diagnosis confirms renal cell carcinoma.

Pathology

1. A 2-year-old boy has a right nephrectomy following an automobile accident for a shattered kidney with uncontrollable bleeding. The histology depicted in Fig. 39.1 is reported as showing interstitial nephritis with cysts consistent with juvenile nephronophthisis. The next step in management is to:
    a. have the pathologist reexamine the specimen for evidence of nephrogenic rests.
    b. have the pathologist reexamine the specimen for an associated teratoma.
    c. image the contralateral kidney for a renal mass.
    d. image the liver for evidence of hepatic fibrosis.
    e. inform the family that the child must be followed carefully for hypertension and decreased renal function.

Answers

1. **b. Tuberous sclerosis and angiomyolipoma.** Angiomyolipomas occur in 40% to 80% of patients with tuberous sclerosis.
2. **a. Hypertension. Hypertension and its sequelae (headache, hypertensive encephalopathy, retinopathy, etc.) are the hallmarks of Ask-Upmark kidney.** Segmental vascular anomalies have been cited as a possible cause of the hypertension, but there is no evidence that renal artery intimal disease is associated. This disease is primarily found in young women and girls.
3. **b. Duration of renal failure.** At first, ARCD was thought to be confined to patients receiving hemodialysis. However, it shortly became apparent that the disorder is almost as common in patients receiving peritoneal dialysis and that it may develop in patients with chronic renal failure who are being managed medically without any type of dialysis. **Thus ARCD appears to be a feature of end-stage kidney disease, rather than a response to dialysis.**
4. **e. b and c.** The most severe form of ARPKD appears earliest in life, in the newborn period. **All patients with ARPKD have liver involvement in the form of hepatic fibrosis and vary in the degree of biliary ectasia and periportal fibrosis.** In both fetus and

newborn, ultrasonography identifies bilateral, very enlarged, diffusely echogenic kidneys, especially when compared with the echogenicity of the liver. The increased echogenicity is due to the presence of numerous microcysts (created by tightly compacted, dilated collecting ducts) that result in innumerable interfaces. Compared with normal newborn kidneys, in ARPKD the pyramids are hyperechogenic because they blend in with the rest of the kidney, and the kidneys typically have a homogeneous appearance.

5. **c. An imbalance of the secretory and absorptive properties in proliferating tubular epithelial cells.** The fundamental processes that are essential for the development and progressive enlargement of renal cysts include (1) proliferation of epithelial cells in segments of renal tubule, (2) accumulation of fluid within the expanding tubule segment, and (3) disturbed organization and metabolism of the extracellular matrix. An imbalance of the secretory and absorptive properties in proliferating epithelial cells leads to a net accumulation of fluid in otherwise normal renal tubules. Recent evidence indicates that, beyond the loop of Henle, tubule cells have the capacity to secrete solutes and fluid on stimulation with 3′,5′-cyclic adenosine monophosphate (cAMP). This secretory flux operates in competition with the more powerful mechanism by which sodium (Na^+) is reabsorbed through apical epithelial Na^+ channels (ENaC). Under conditions in which Na^+ reabsorption is diminished, the net secretion of sodium chloride (NaCl) and fluid occurs.
6. **a. The genetic defect is located on the short arm of chromosome 16.** Infants with ARPKD have hepatic fibrosis, and infants with ADPKD rarely have hepatic fibrosis but commonly have cysts in the liver. Renal cysts are frequently seen in individuals on ultrasonography by the age of 20 years. Glomerular cysts are sometimes found in the kidneys of newborns diagnosed with ADPKD. The risk of renal cell carcinoma in patients with ADPKD is no higher than that in the general population.
7. **c. Cerebellar hemangioblastomas.** All are extrarenal manifestations of ADPKD except cerebellar hemangioblastomas,

which are seen in patients with von Hippel-Lindau disease.

8. **e. Multicystic dysplastic kidneys appear more often in females and more often on the right side.** At any age, the condition is more likely to be found on the left side. Males are more likely to have unilateral multicystic dysplastic kidneys (2.4:1).
9. **a. Bleeding into a cyst.** Pain (flank and/or abdominal) is the most common presenting symptom in adults. This results from a number of possible factors: mass effect (cysts impinging on abdominal wall or neighboring organs), bleeding into the cysts, urinary tract infection (including infected cysts), and nephrolithiasis.
10. **e. VHL.** The gene associated with the transmission of von Hippel-Lindau disease is located on chromosome 3. In non–von Hippel-Lindau patients with sporadic clear cell renal cell carcinoma, 50% of cell lines are associated with a mutational form of the VHL gene.
11. **a. In males younger than 4 years and in females older than 30 years.** The great majority of patients present before the age of 4 years or after the age of 30 years. Five percent present between 4 and 30 years. The patient is 2 times as likely to be male if younger than 4 years and 8 times as likely to be female if older than 30 years.
12. **b. NPH is an autosomal recessive disorder, whereas MCKD is an autosomal dominant disease. Although either condition can occur sporadically, juvenile nephronophthisis usually is inherited as an autosomal recessive trait, whereas medullary cystic disease usually is inherited in an autosomal dominant fashion. Juvenile nephronophthisis and medullary cystic disease both cause polydipsia and polyuria in more than 80% of cases, but not to the extent observed in patients with diabetes insipidus. Pathologically, NPH and MCKD are similar.** Histologically, there is a characteristic triad present that includes (1) irregular thickening and disintegration of the tubular basement membrane, (2) marked tubular atrophy with cyst development, and (3) interstitial cell infiltration with fibrosis. Twenty percent of juvenile nephronophthisis families have extrarenal manifestations, whereas MCKD usually affects only the kidneys. Another

important difference between the two entities is that renal failure develops in patients with NPH at a mean age of 13 years and almost always before 25 years. MCKD is a milder disease when it presents in early adulthood, but it will manifest in all patients by 50 years (Bernstein and Gardner, 1979). [a] End-stage renal disease (ESRD) in patients with MCKD most often develops in the third or fourth decade of life.

13. **c. von Hippel-Lindau disease. Tuberous sclerosis and von Hippel-Lindau disease are associated with epithelial hyperplasia (and adenomas as well) and have an increased incidence of renal cell carcinoma (tuberous sclerosis, 2%, and von Hippel-Lindau disease, 35% to 38%).**
14. **d. Data from large series show that MCDK is associated with an increased risk for hypertension. All statements are true of MCDK, except that large series indicate MCDK is NOT associated with an increased risk of hypertension.**
15. **c. Renal angiomyolipoma and cardiac rhabdomyoma. Definitive diagnosis of tuberous sclerosis (TSC) is dependent on the presence of certain major and minor clinical features.** The diagnosis of TSC requires two major features (renal angiomyolipoma, facial angiofibromas or forehead plaques, nontraumatic ungual or periungual fibroma, three or more hypomelanotic macules, shagreen patch, multiple retinal nodular hamartomas, cortical tuber, subependymal nodule, subependymal giant cell astrocytoma, cardiac rhabdomyoma, lymphangioleiomyomatosis) or one major plus two minor features (multiple renal cysts, nonrenal hamartoma, hamartomatous rectal polyps, retinal achromic patch, cerebral white matter radial migration tracts, bone cysts, gingival fibromas, “confetti” skin lesions, multiple enamel pits).
16. **e. Renal cell carcinomas, the most common manifestation, are seen in the vast majority of patients.** All statements are true of VHL disease except that renal cysts, NOT *renal cell carcinoma*, are the most common and often earliest manifestation as seen in 76% of patients.

17. **d. Lymphatic system.** The predominant type of renal sinus cyst appears to be one derived from the lymphatics.
18. **e. Resolve before birth.** In 28 of 11,000 fetuses with renal cysts, 25 fetuses had the cysts resolve before birth. Of two cysts that remained postnatally, in one it was the first sign of a multicystic kidney.
19. **c. 33%.** In adults, the frequency of renal cyst occurrence increases with age. Using CT, one group demonstrated a 20% incidence of cysts by 40 years and approximately 33% incidence of cysts after 60 years.
20. **a. Trimethoprim-sulfamethoxazole, chloramphenicol, fluoroquinolones. In the experience of one group of researchers, the only dependable antibiotics were those that were lipid soluble, namely, trimethoprim-sulfamethoxazole and chloramphenicol. Chloramphenicol produced better results. The fluoroquinolones, which are also lipid soluble, are proving useful.** If a patient with suspected pyelonephritis does not respond to an antibiotic, and if the antibiotic used is not lipid soluble, one must consider whether the infection may be present in a noncommunicating cyst.
21. **b. Avoidance of surgical treatment for large or multiple cysts in patients with chronic flank pain.** All are reasonable treatment strategies for a patient with ADPKD, except that when conservative measures of chronic pain treatment fail, surgical management must be considered. Ultrasonography- or CT-guided cyst aspiration is a straightforward procedure and may be both diagnostic and therapeutic. Surgical unroofing of multiple or very large cysts can potentially alleviate symptoms of pain and can be performed either laparoscopically or through open flank or dorsal lumbotomy incisions. Surgical intervention appears to only improve symptomatology and does not appear to either accelerate the decline of renal function or preserve declining renal function.
22. **b. 18% to 43%.** Contralateral vesicoureteral reflux is seen even more often than contralateral ureteropelvic junction obstruction, being identified in 18% to 43% of infants.

23. **d. ARCD.** The most common presentation of ARCD is loin pain, hematuria, or both. Bleeding occurs in as many as 50% of patients.
24. **b. The cysts have a haphazard distribution; there is absence of a central or medial large cyst; and there are no obvious communications between the cysts.** Renal masses in infants most often represent either multicystic kidney disease or hydronephrosis, and it is important to distinguish the two, especially if the surgeon wishes to remove a nonfunctioning hydronephrotic kidney or repair a ureteropelvic junction obstruction while leaving a multicystic organ in situ. In newborns, ultrasonography is generally the first study performed. **In a few cases, it is difficult to distinguish multicystic kidney disease from severe hydronephrosis. In general, however, the multicystic kidney has a haphazard distribution of cysts of various sizes without a larger central or medial cyst and without visible communications between the cysts.** Frequently, very small cysts appear in between the large cysts. **By comparison, in ureteropelvic junction obstruction, the cysts or calyces are organized around the periphery of the kidney, connections can usually be demonstrated between the peripheral cysts and a central or medial cyst that represents the renal pelvis, and there is an absence of small cysts between the larger cysts.** When there is an identifiable renal sinus, the diagnosis is more likely to be hydronephrosis than multicystic kidney.
25. **a. Benign multilocular cyst.** For the benign multilocular cystic lesion, certain authors prefer the term *cystic nephroma*, because this term implies a benign but neoplastic lesion.
26. **a. ARPKD and congenital hepatic fibrosis.** All patients with ARPKD have varying degrees of congenital hepatic fibrosis.
27. **b. ADPKD and chromosomes 4 and 16.** For the genetic cystic disease ADPKD, the chromosomal defect is on chromosome 16 for *PKD1* and 4 for *PKD2; PKD3* has not been mapped. Autosomal recessive polycystic kidney disease involves chromosome 6; tuberous sclerosis involves chromosomes 9 and 16; von Hippel-Lindau disease involves chromosome 3; and juvenile nephronophthisis involves chromosome 2.

28. **c. II F.** The Bosniak classification has recently been updated to include category II F. Bosniak I—simple cyst with imperceptible wall, rounded. Workup: none. Percentage malignant: ~0%. Bosniak II—minimally complex with a few thin <1 mm septa or thin calcifications (thickness not measurable); non-enhancing high-attenuation (due to proteinaceous or hemorrhagic contents) renal lesions <3 cm. These lesions are generally well marginated. Workup: none. Percentage malignant: ~0%. Bosniak IIF—minimally complex with increased number of septa, minimally thickened with nodular or thick calcifications. There may be perceived (but not measurable) enhancement of hairline-thin smooth septa, minimal thickening of the wall with perceivable enhancement, hyperdense cyst >3 cm diameter, mostly intrarenal (less than 25% of wall visible); no enhancement. Requires follow-up (the "F" in 2F is for "follow"): needs ultrasound/CT/MRI follow-up—no strict rules on the time frame but reasonable at 6 months. Percentage malignant: ~5%. Bosniak III—indeterminate with thick, nodular multiple septa or wall with measurable enhancement. Treatment/workup: partial nephrectomy. Percentage malignant: ~55%. Bosniak IV—clearly malignant with solid mass with a large cystic or a necrotic component. Treatment: partial or total nephrectomy. Percentage malignant: ~100%
29. **d. The presence of numerous microcysts created by tightly compacted, dilated collecting ducts that result in innumerable ultrasonographic interfaces.** In both fetuses and newborns with ARPKD, ultrasonography identifies bilateral, very enlarged, diffusely echogenic kidneys, especially when compared with the echogenicity of the liver. The increased echogenicity is due to the presence of numerous microcysts (created by tightly compacted, dilated collecting ducts) that result in innumerable interfaces. Compared with normal newborn kidneys, in ARPKD the pyramids are hyperechogenic because they blend in with the rest of the kidney, and the kidneys typically have a homogeneous appearance.
30. **b. Thickness of cyst wall less than or equal to 3 mm.** One can safely make the diagnosis of a classic benign simple cyst by ultrasonography when the following criteria are met: (1) absence of

internal echoes; (2) sharply defined, thin, distinct wall with a smooth and distinct margin; (3) good transmission of sound waves through the cyst with consequent acoustic enhancement behind the cyst; and (4) spherical or slightly ovoid shape. The CT criteria for a simple cyst are similar to those used in ultrasonography: (1) sharp, thin, distinct, smooth walls and margins; (2) spherical or ovoid shape; and (3) homogeneous content.

31. **b. Bilateral upper pole partial nephrectomy.** Because the tumors that characterize VHL disease are frequently multiple, bilateral, and recurrent, close surveillance and minimization of surgical procedures constitute the mainstay of treatment. For patients who have VHL disease and all patients who have hereditary cancer syndromes, the goal of treatment is cancer control, not cancer cure, and preservation of functional parenchyma to avoid the morbidity associated with renal loss. In patients who have VHL disease, surgical resection is performed with the understanding that microscopic disease probably is left behind. Currently, nephron-sparing surgery should be considered the standard of care for treating low-grade renal cell carcinoma in the setting of VHL disease. Patients with high-grade disease are still probably best served with bilateral nephrectomy. Despite having the objective of sparing as much renal parenchyma as possible and preventing metastasis of the lesions already present, it is not curative surgery (Reed, 2009). Although this approach does not reduce the risk of recurrence, reported to be 75% to 85%, the 10-year disease-specific survival rates are quite high (81% to 94%) (Malek, 1987; Steinbach, 1995; Roupret, 2003; Ploussard, 2007). Classically, the survival rate after nephrectomy has been only 50%. Because most of these tumors are low grade, a nephron-sparing approach provides very good survival rates while avoiding the diminished quality of life that comes with bilateral nephrectomy and subsequent dialysis/transplantation. Laparoscopic and percutaneous image-guided ablative techniques, such as radiofrequency ablation and cryoablation, have also been used and are currently under investigation.

Pathology

1. **e. Inform the family that the child must be followed carefully for hypertension and decreased renal function.** The histology is classic for juvenile nephronophthisis showing chronic interstitial nephritis-fibrosis, atrophic tubules, and glomerular microcysts. It is usually inherited as an autosomal recessive. These patients have a high likelihood of developing hypertension and eventually chronic renal failure. They may also develop retinitis pigmentosa.

Chapter review

1. Potter facies is manifested by hypertelorism, prominent inner canthal folds, and a recessive chin.
2. Dysplasia is histologically manifested by embryonic, immature mesenchyme, and primitive renal components; it is often associated with ureteric bud abnormalities and/or urinary obstruction.
3. Renal hypoplasia is manifested by less than the normal number of calyces and nephrons with absence of dysplasia.
4. Oligomeganephronia is a reduced number of nephrons with hypertrophy of the remaining nephrons. Many patients with the disorder develop renal failure.
5. The Ask-Upmark kidney (segmental hypoplasia) is often associated with reflux and patients develop severe hypertension.
6. Autosomal dominant polycystic kidney disease is associated with cysts of the liver, pancreas, spleen, and lungs, berry aneurysms; colonic diverticula; aortic aneurysms; and mitral valve prolapse. It usually becomes clinically manifest in the fourth and fifth decades.
7. Benign multilocular cyst and cystic nephroma fall into a spectrum of disease, with multilocular cyst on the one end being benign and, on the other end of the spectrum, cystic Wilms tumors being malignant. Multilocular cyst with nodules of Wilms tumor lies in the middle.
8. In the adult, there is a multilocular cystic renal cell carcinoma. Multilocular cystic lesions should therefore be removed.

9. One-third of patients with medullary sponge kidneys have hypercalciuria.
10. Congenital bilateral absence of the vas is associated with cystic fibrosis; unilateral absence of the vas may be associated with ipsilateral absence of the kidney.
11. Von Hippel-Lindau syndrome is inherited as an autosomal dominant and is manifested by cerebellar and retinal hemangioblastomas; cysts of the pancreas, kidney, and epididymis; epididymal cystadenoma, pheochromocytoma, and clear cell renal cell carcinoma.
12. Multicystic dysplastic kidney has no identifiable normal renal parenchyma; it is not associated with an increased risk of either hypertension or malignancy.
13. Medullary sponge kidney has dilated collecting ducts and is associated with hypercalciuria, hypocitraturia, and renal calculi.
14. All patients with autosomal recessive polycystic kidney disease (ARPKD) have liver involvement in the form of hepatic fibrosis and varying degrees of biliary ectasia and periportal fibrosis.
15. Juvenile nephronophthisis usually is inherited as an autosomal recessive trait, whereas medullary cystic disease usually is inherited in an autosomal dominant fashion. Juvenile nephronophthisis and medullary cystic disease both cause polydipsia and polyuria, and pathologically they are similar.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition*, on the Expert Consult website.

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Pathophysiology of Urinary Tract Obstruction

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Keywords

congenital urinary obstruction; hydronephrosisrenal dysplasia; renal fibrosis; urinary tract obstruction

Questions

1. Congenital obstruction differs from acquired obstruction in that it:
    a. causes renal atrophy.
    b. induces interstitial fibrosis.
    c. alters renal homeostasis.
    d. affects tubular function.
    e. affects glomerular development.
2. Renal dysplasia associated with obstruction is characterized by:
    a. renal atrophy.
    b. glomerular cysts.
    c. fibromuscular collars.
    d. heterotopic bone formation.
    e. excess production of afferent arteriole renin.
3. After relief of a unilateral obstructing lesion, decreasing relative uptake on radionuclide renal imaging is most likely due to:
    a. glomerular hyperfiltration.
    b. asymmetrical renal growth.

c. established renal tubular fibrosis.
d. neural imbalance.
e. compensatory hypertrophy.

4. In the obstructed kidney, epidermal growth factor (EGF) has been shown to:
   a. reduce renal apoptosis.
   b. reduce glomerular sclerosis.
   c. accelerate interstitial fibrosis.
   d. improve collecting duct function.
   e. reduce renin recruitment in the afferent arteriole.
5. Epithelial to mesenchymal transformation in the developing kidney is:
   a. a one-way process.
   b. the basis for glomerular sclerosis.
   c. integral to glomerular development.
   d. seen only in the setting of obstruction.
   e. reflected in the presence of α-smooth muscle actin.
6. Regulation of the extracellular matrix in the kidney:
   a. depends on normal expression of EGF.
   b. depends entirely on collagen synthesis.
   c. is not related to angiotensin expression.
   d. depends on balanced synthesis and degradation.
   e. is independent of transforming growth factor-β (TGF-β) activity.
7. The principal effects of congenital renal obstruction are:
   a. glomerulosclerosis, interstitial fibrosis, and atrophy.
   b. hypoplasia and increased epithelial-mesenchymal transformation.
   c. altered growth regulation, renal differentiation, and functional integration.
   d. glomerulosclerosis, renin downregulation, and tubular hypertrophy.
   e. increased growth, fibrosis, and tubular atrophy.
8. In the fetal kidney, angiotensin activity:
   a. is tightly regulated by EGF.
   b. acts predominantly through the AT-1 receptor.

c. affects epithelial-mesenchymal transformation.
d. is an important regulator of renal growth.
e. is unaffected by renal obstruction.

9. Inflammatory change in the congenitally obstructed kidney:
    a. is similar to those seen in postnatally obstructed kidneys.
    b. is mediated by the renin-angiotensin system.
    c. is minimal in the absence of overt infection.
    d. is the key element in glomerular damage.
    e. affects renal interstitial fibrosis.

Answers

1. **e. Affects glomerular development.** Only congenital obstruction will change glomerular development, whereas acquired obstruction can produce all of the other changes indicated. Because it occurs during development, congenital obstruction can produce an altered developmental pattern, whereas acquired obstruction cannot change an already established pattern, only damage or distort it.
2. **c. Fibromuscular collars.** One of the histologic hallmarks of dysplasia is fibromuscular collars, so-called *primitive ducts* reflecting abnormal differentiation of the peritubular mesenchyme. Renal growth impairment is common with dysplasia; however, this is not atrophy but growth failure. Glomerular cysts are not characteristic of dysplasia. Heterotopic cartilage may be seen, but not bone. Excess renin expression may be seen in obstruction without dysplasia.
3. **b. Asymmetrical renal growth.** When renal function appears to decline after relief of obstruction, it is often due to different growth and functional development rates of the two kidneys, when the affected kidney cannot increase its absolute function as rapidly as the other intact kidney. This produces a progressive differential functional uptake on nuclear imaging that gives the impression of functional loss that is relative and not absolute.
4. **a. Reduce renal apoptosis.** Administration of EGF to the congenitally obstructed kidney can reduce renal apoptosis and

reduce the effects of growth impairment. The other effects have not been reported.

5. **e. Reflected in the presence of α-smooth muscle actin.** Epithelial to mesenchymal transformations are an important part of renal development but have not been shown to be part of normal glomerular development. It is the presumed basis for the presence of α-smooth muscle actin in the obstructed kidney. It is bidirectional.
6. **d. Depends on balanced synthesis and degradation.** Extracellular membrane (ECM) regulation is due to collagen synthesis rates as well as to the rate of ECM breakdown. The latter is determined by the balanced activities of the tissue inhibitors of metalloproteinases (TIMPs) and matrix metalloproteinases (MMPs); these are regulated in part by TGF-β and the renin-angiotensin system.
7. **c. Altered growth regulation, renal differentiation, and functional integration.** The key patterns defining congenital renal obstruction are altered growth regulation, renal differentiation, and functional integration, although interstitial fibrosis, tubular hypotrophy, and increased epithelial-mesenchymal transformation are components of these changes.
8. **d. Is an important regulator of renal growth.** In the developing kidney, angiotensin is an important growth regulator, as well as mediator of fibrosis, and is altered significantly by obstruction. Fetal angiotensin acts predominantly through the AT2 receptor until late in gestation, when the AT1 receptor begins to exert a greater role.
9. **c. Is minimal in the absence of overt infection.** In contrast to acquired obstruction, congenital obstruction is not characterized by a significant inflammatory infiltrate, except when complicated by infection.

Chapter Review

1. A damaged kidney does not have the functional reserve to maintain normal renal function over time as the child grows. Thus, with time,

the creatinine concentration will rise in patients who have no functional reserve in their nephron mass.

2. Obstructive processes may produce dysplasia.
3. In some cases, congenitally obstructed kidneys are smaller. This is not a result of atrophy but of hypoplasia.
4. A critical determinant of dysplasia in the kidney is complete obstruction early in gestation.
5. A universal characteristic of obstruction is renal fibrosis with infiltration of the interstitium with extracellular matrix.
6. The frequency with which a post-obstructed kidney has progressive deterioration in renal function is 20%–40%; it does not have the renal reserve of a normal kidney and may have a reduced number and function of nephrons. This results in hyperfiltration resulting in glomerulosclerosis and a progressive decline in renal function.
7. Measures of urinary sodium, chloride, osmolality, and calcium correlate with fetal renal function. When these measures approach serum levels, irreversible damage is suggested.
8. The distinctness of the calyx is helpful in determining the functionality of the obstruction. Thus, no caliectasis suggests a mild to minimal degree of functional obstruction.
9. ^{99m}Tc-Mercaptoacetyltriglycine is generally used for diuretic renography. It is taken up and excreted by the renal tubules relatively rapidly with little glomerular filtration.
10. Though other factors may play a role, if the diuretic renogram shows a T1/2 greater than 20 minutes, the kidney is presumed to be obstructed; if it is less than 10 minutes, it is presumed normal; if it is between 10 and 20 minutes, it is considered indeterminate.
11. Obstruction can be very harmful to the developing kidney, far beyond that seen in the mature kidney. The effects of obstruction may not be reversible owing to alterations in structure and function in the developing kidney that do not occur in the mature kidney.
12. When obstruction results in a growth impairment, the results are fewer nephron units and delayed nephron maturation.
13. Angiotensin appears to be a key modulator of the inflammatory response in renal obstruction. It is also an important growth

regulator, as well as mediator of fibrosis, and is altered significantly by obstruction.

14. Renal renin is increased in the obstructed kidney.
15. Inflammation with interstitial infiltration of lymphocytes is a common finding in postnatal acquired obstruction.
16. When renal function appears to decline after relief of obstruction, it is often due to differential growth and functional development rates of the two kidneys, when the affected kidney cannot increase its absolute function as rapidly as the contralateral intact kidney. This produces a progressive differential functional uptake on nuclear imaging that gives the impression of functional loss that is relative and not absolute.
17. In contrast to acquired obstruction, congenital obstruction is not characterized by a significant inflammatory infiltrate, except when complicated by infection.